

Duality in SEK

$$(P): \max \{ c^T x : Ax = b, x \geq 0 \} \quad \text{"primal"}$$

$$(D): \min \{ b^T y : A^T y \geq c \} \quad \text{"dual"}$$

Example:

$$(P) \quad \max (1 \ 1 \ 0)x$$

$$\text{s.t.} \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} x = \begin{pmatrix} 8 \\ 17 \end{pmatrix} \quad \begin{matrix} \rightarrow y_1 \\ \rightarrow y_2 \end{matrix}$$

$x \geq 0$

dual var

$$(D) \quad \min (8 \ 17)y$$

$$\text{s.t.} \quad \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} y \geq \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \begin{matrix} \rightarrow x_1 \\ \rightarrow x_2 \\ \rightarrow x_3 \end{matrix}$$

primal var

Note: Each primal constraint corresponds to a dual variable, each primal variable corresponds to a dual constraint.

Weak Duality Theorem: If $\bar{x}, \bar{y}$ are feasible for (P), (D), then $c^T \bar{x} \leq b^T \bar{y}$. Moreover, if $c^T \bar{x} = b^T \bar{y}$, then $\bar{x}, \bar{y}$ are optimal for (P), (D).

Example: $\bar{x} = (0, 1, 2)^T$ is feasible for (P) above, obj value 1. $\bar{y} = (1, 0)^T$ is feasible for (D), obj value 8 > 1. But $\hat{y} = (-2, 1)^T$ is feasible for (D) with obj value 1. So $\bar{x}, \hat{y}$ are optimal for (P), (D).

Quick proof sketch:

$$c^T \bar{x} \leq (A^T \bar{y})^T \bar{x} = \bar{y}^T (A \bar{x}) = \bar{y}^T b = b^T \bar{y}.$$

$\uparrow$
 $c \leq A^T \bar{y}, x \geq 0$
 $\uparrow$
 $Ax = b$

Corollary: For a primal-dual pair, if one is unbounded, then the other is infeasible.

Note: The converse is not true. It is possible that both are infeasible.

Also, weak duality theorem does not guarantee that optimal values are equal.

Strong duality theorem: (P) has an optimal solution if and only if (D) has an optimal solution, and they have the same optimal value.

Sketch proof: Run simplex on (P). $y = A_B^{-T} c_B$ in the final basis is optimal for (D).

Certificate of optimality: To show that $\bar{x}$ is optimal for (P), provide a certificate $\bar{y}$ and check: ① $\bar{x}$ feasible for (P); ② $\bar{y}$ feasible for (D); and ③ $c^T \bar{x} = b^T \bar{y}$.

Duals of other forms

$$\begin{array}{ll} \max & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0 \end{array} \quad \longleftrightarrow \quad \begin{array}{ll} \min & b^T y \\ \text{s.t.} & A^T y \leq c \\ & y \geq 0 \end{array}$$

Follow the conversion table:

max	min
$\leq$ constraints	≥ 0 variables
$\geq$ "	≤ 0 "
$=$ "	free "
≥ 0 variables	$\geq$ constraints
≤ 0 "	$\leq$ "
free "	$=$ "

If the primal is a minimization problem, apply the table in $\leftarrow$ direction.

Example:

$$\begin{array}{ll} \max & (3 \ -4 \ 1)x \\ \text{s.t.} & \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix} x \leq \begin{pmatrix} 5 \\ -3 \end{pmatrix} \\ & x_1 \geq 0 \quad x_2 \leq 0 \quad x_3 \text{ free} \end{array} \quad \rightarrow \quad \begin{array}{ll} \min & (5 \ -3)y \\ \text{s.t.} & \begin{pmatrix} 3 & 1 \\ -1 & 2 \\ 2 & 3 \end{pmatrix} y \geq \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix} \\ & y_1 \geq 0 \quad y_2 \leq 0 \end{array}$$

(The dual of (D) is (P).)

Note: Weak and Strong duality theorems hold for LPs in general.